not be replaced, the air would clear, and the surface would become visible in ordinary light. It would become a world very like Triton.

These two worlds are not identical. The interior of Titan seems to contain much more ice than that of Triton, and much less rock. Titan's diameter is almost twice that of Triton. Still, if placed at the same distance from the Sun they would look like sisters. Alan Stern of the Southwest Research Institute suggests that they are two members of a vast collection of small worlds rich in nitrogen and methane that formed in the early Solar System. Pluto, yet to be visited by a spacecraft, appears to be another member of this group. Many more may await discovery beyond Pluto. The thin atmospheres and icy surfaces of all these worlds are being irradiated —by cosmic rays, if nothing else—and nitrogen-rich organic compounds are being formed. It looks as if the stuff of life is sitting not just on Titan, but throughout the cold, dimly lit outer reaches of our planetary system.

Another class of small objects has recently been discovered, whose orbits take them—at least part of the time—beyond Neptune and Pluto. Sometimes called minor planets or asteroids, they are more likely to be inactive comets (with no tails, of course; so far from the Sun, their ices cannot readily vaporize). But they are much bigger than the run-of-the-mill comets we know. They may be the vanguard of a vast array of small worlds that extends from the orbit of Pluto halfway to the nearest star. The innermost province of the Oort Comet Cloud, of which these new objects may be members, is called the Kuiper Belt, after my mentor Gerard Kuiper, who first suggested that it should exist. Short-period comets—like Halley's—arise in the Kuiper Belt, respond to gravitational tugs, sweep into the inner part of the Solar System, grow their tails, and grace our skies.

Back in the late nineteenth century, these building blocks of worlds—then mere hypotheses—were called "planetesimals." The flavor of the word is, I suppose, something like that of "infinitesimals": You need an infinite number of them to make anything. It's not quite *that* extreme with planetesimals, although a very large number of them would be required to make a planet. For